

Adonay Tesfamichael

IT-214-006

Hierarchy Design

This is a database that describes a disjoint hierarchy of medical company. This ERD has 4 entities, which explains an employee information, each employee has a type of role, like being a doctor, nurse or a staff.

The main entities are:

- **EMPLOYEE:** HAVE TYPE OF EMPLOYEE ROLES, (DOCTOR, NURSE AND STAFF)
 - **Surrogate PK:** ID_AT

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
NAME	VARCHAR(50)	COMPOSITE	SINGLE	REQUIRED	DUPLICATED
FIRST_NAME_AT	VARCHAR(25)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
LAST_NAME_AT	VARCHAR(25)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SSN_AT	VARCHAR(11)	SIMPLE	SINGLE	REQUIRED	UNIQUE
PHONE_AT	CHAR(12)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
TYPE	CHAR(1)	SIMPLE	SINGLE	REQUIRED	DUPLICATED

- **DOCTOR:**
 - **Surrogate PK:** N/A
 - **FK:** EMPLOYEE_AT_ID_AT references the pk ID_AT in the entity EMPLOYEE_AT

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
EMAIL_AT	VARCHAR(25)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SPECIALTY_AT	VARCHAR(20)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
MEDICAL_SCHOOL_AT	VARCHAR(30)	SIMPLE	SINGLE	REQUIRED	DUPLICATED

- **NURSE:**
 - **Surrogate PK:** N/A
 - **FK:** EMPLOYEE_AT_ID_AT references the pk ID_AT in the entity EMPLOYEE_AT

DOCTOR

Name: DOCTOR_AT					
Column	Datatype	PK	NN	UQ	
EMAIL_AT	VARCHAR(25)	☐	☒	☐	
SPECIALTY_AT	VARCHAR(20)	☐	☒	☐	
MEDICAL_SC...	VARCHAR(30)	☐	☐	☐	
EMPLOYEE_A...	INT	☒	☒	☐	

NURSE

Name: NURSE_AT					
Column	Datatype	PK	NN	UQ	
PROGRAM_C...	VARCHAR(30)	☐	☒	☐	
EMPLOYEE_A...	INT	☒	☒	☐	

STAFF

Name: STAFF_AT					
Column	Datatype	PK	NN	UQ	
ROLE_AT	CHAR(21)	☐	☒	☐	
EMPLOYEE_A...	INT	☒	☒	☐	